

Multiple-chain preorder polytopes: two proofs of the exponential formula

A rectangular extension, unequal block sizes, algebraic generating functions,
fixed-support polynomiality, and support asymptotics

Research draft prepared for Vladimir Reshetnikov

20 September 2026

Abstract

Let $H_{n,d}(t)$ enumerate the nonnegative integer points satisfying $\sum_{i \leq r} \sum_{j \leq d} a_{ij} \leq dr$ for $1 \leq r \leq n$, with t marking nonzero coordinates. We give two complete and structurally different formal-power-series proofs of

$$\sum_{n \geq 0} H_{n,d}(t) z^n = \exp \left(\sum_{m \geq 1} \frac{z^m}{m} \sum_{j=0}^{dm} \binom{dm}{j}^2 t^j \right).$$

This is the conjectural multiple-chain identity of Athanasiadis and Chapoton, Section 9.5 of arXiv:2605.26916v1, with its normalization removed. The first proof encodes lattice points by two subsets and applies a constant-term excursion factorization with a bounded Laurent-polynomial symbol; the second runs the original coordinate slack process, whose symbol is an unbounded Laurent series, and evaluates the positive part by stars and bars. Replacing the budget dr by cr gives the stronger bridge polynomial $\sum_j \binom{dm}{j} \binom{cm}{j} t^j$; allowing unequal block sizes gives a commutative formula summed over orders, which is sharp because the support polynomial is not a function of the multiset of block sizes. We derive a support-preserving rectangular duality, recurrences, Ehrhart evaluations, a root-of-unity product with two degree bounds, an explicit quartic for double chains, and a gamma-coefficient identity with both a block-boundary path model and an algebraic constant-term derivation. For each fixed support size j the coefficient is a polynomial in n of exact degree $2j$ with leading coefficient $d^{2j}/(j!(j+1)!)$. We obtain a Stieltjes moment representation in the chain-length variable, explicit $n^{-3/2}$ asymptotics with two expressions for the boundary constant, and a central limit theorem for support size including an exact variance formula and its constant-order correction. Two independent exact verifier suites, over disjoint parameter ranges, accompany the article.

Status and scope. The main identity is proved for every integer $d \geq 1$, not just a finite range. This is an AI-assisted research draft, not an independently refereed or proof-assistant-verified paper; neither its correctness nor its priority has been certified. The selected identity is stated as experimental in the retrieved source; a targeted literature search found no subsequent proof, but this is a bounded check and not a certification of publication priority. The walk factorization is classical Spitzer/Wiener–Hopf background and the earlier gamma-positivity theorem for composition polytopes is explicitly credited; neither is claimed as new here.

Contents

1	The selected problem and its precise meaning	4
1.1	Relation to the supplied manifest	4
1.2	Provenance of this article	4
1.3	Chain preorders, not layers of incomparable elements	4
1.4	What is proved, and what is not claimed	5
1.5	Deliberate redundancies	6
2	A rectangular lattice-point theorem, proved by excursions	7
2.1	Statement	7
2.2	From an array to two subsets	7
2.3	The formal factorization, in two forms	8
2.4	First proof of the theorem	9
3	A second proof by the original slack process	10
3.1	Slack walks and the block weight	10
3.2	The step symbol and its ring	10
3.3	The endpoint refinement	11
3.4	Evaluating the positive part	11
3.5	Why the type- $\mathbb{B}$ polynomial appears	12
4	Duality, recurrences, and exact enumeration	12
4.1	A support-preserving rectangular duality	12
4.2	Convolution and partition formulas	12
4.3	Examples and boundary cases	13
4.4	Uniform dilations and Ehrhart evaluation	14
5	Algebraic generating functions	14
5.1	The one-block-size series, two ways	14
5.2	A root-of-unity product for every block size	15
5.3	An explicit quartic for double chains	16
6	Gamma coefficients: a path model and an algebraic derivation	16
6.1	The block-boundary path model	17
6.2	The algebraic derivation, integrality and strict positivity	18
7	Fixed support size: polynomiality in the number of blocks	18
7.1	The first two coefficients	20
8	A Stieltjes moment representation in the number of blocks	20
9	Unequal block sizes	21
10	Singularity analysis and explicit leading asymptotics	22

11 Support size: exact variance and a central limit theorem	24
12 Reproducibility, proof audit, and remaining questions	26
12.1 Independent checks actually performed	26
12.2 A short audit of the mathematical dependencies	27
12.3 Reproduction commands	27
12.4 What would be a genuinely different next problem?	28
A A compact proof certificate	28
B Exact data and source provenance	29

1 The selected problem and its precise meaning

1.1 Relation to the supplied manifest

The supplied manifest [8] records seventy-one packages and describes two of them on preorder polytopes: *Reciprocity and Matrix Duality for Preorder Polytopes* and *A Reflexive Root-Polytope Model for Preorder h -Polynomials*. Their catalogued targets concern reciprocity, transpose duality, and polytopal realization. The present target is instead the multiple-chain generating function in Section 9.5 of Athanasiadis–Chapoton [1]. This identity is not among the results described in the manifest. No theorem claimed in those two packages is used below; both proofs start from the defining inequalities. The manifest is a selection and duplicate-avoidance aid, not independent evidence that its other proofs are correct; it expressly warns that its statements describe unrefereed claims.

1.2 Provenance of this article

This article is the merger of two independently prepared research packages, both dated 20 September 2026 and both targeting the same unnumbered identity.

- **enumerative-combinatorics/multiple-chain-exponential-formula** (the base of this merge) proved the identity by encoding each lattice point as a pair of subsets with a prefix-domination condition, so that the admissible arrays become nonnegative *excursions* whose one-block symbol is the bounded Laurent polynomial $(1+u)^c(1+t/u)^d$; the logarithm of the excursion series is then the constant term $\text{CT}_u P^m$, evaluated by the binomial theorem. It proved the statement in the rectangular generality (d, c) recorded as Theorem 2.1 below, and it carried the rectangular duality, the Ehrhart evaluation, the Stieltjes moment section, and the exact variance with its constant-order correction. It also contained, as an appendix, a rectangular form of the second proof.
- **enumerative-combinatorics/multiple-chains-exponential-identity** proved the identity by running the original coordinate slack process: the block slack walk has step series $S_k(u) = u^k(1+q/(u-1))^k$ expanded at $u = \infty$, the admissible arrays are *meanders* of that walk, the positive-part factorization is carried out in $\mathbb{Q}[q]((u^{-1}))[[z]]$, and $u = 1$ is substituted only after the projection, where stars and bars produces the type-B Narayana polynomial. It proved the diagonal case $c = d$ only, but it carried the fixed-support polynomiality theorem, the mixed-block-size theorem with its order-dependence counterexample, the Lagrange-inversion Narayana row, the gamma integrality and strict-positivity arguments, and a verifier suite over a deeper range of parameters.

In the merged notation below, the second package's k is this article's d and its q is this article's t ; its $A_k(q)$ is renamed $K_d(t)$ to avoid a collision with the array-counting symbol $A_{n;d,c}(t)$, its $\mathcal{N}(q, z)$ is this article's $g(w, t)$, and its $P_{\geq 0}$ and $P_{< 0}$ are written $[\cdot]_{\geq 0}$ and $[\cdot]_{< 0}$. Both packages removed the source's Laurent normalization in the same way, and their shared numerical data agree exactly. The second proof of Section 3 is not a paraphrase of the first: it is a different walk, with a different symbol, in a different ring, under a different boundary condition.

1.3 Chain preorders, not layers of incomparable elements

For $d \geq 1$ and $n \geq 0$, partition a ground set of size dn into ordered blocks $B_1, \dots, B_n$, each of size d . Within a block all elements are equivalent under the preorder, and its quotient order is $B_1 < \dots < B_n$. The nonempty order ideals are therefore exactly the complete prefixes $B_1 \cup \dots \cup B_r$. In particular, these are *not* ordinal sums of d -element antichains, whose ideals

would allow partial blocks. A vertex of the quotient poset represents an entire equivalence class, so the preorder $\tau_{n,d}$ has n quotient vertices but dn underlying elements, and its polytope has dimension dn , not n .

The associated polytope and its support enumerator are

$$Q_{n,d} = \left\{ a \in \mathbb{R}_{\geq 0}^{dn} : \sum_{i=1}^r \sum_{j=1}^d a_{ij} \leq dr \quad (1 \leq r \leq n) \right\}, \quad (1.1)$$

$$H_{n,d}(t) = \sum_{a \in Q_{n,d} \cap \mathbb{Z}^{dn}} t^{\#\{(i,j): a_{ij} > 0\}}, \quad H_{0,d}(t) = 1. \quad (1.2)$$

We use $\mathbb{N} = \{0, 1, 2, \dots\}$. These are the source's polynomials $h(\tau_{n,d}, t)$ [1, Sections 1, 5, and 9.5], and (1.1) is the composition-polytope model of [2, Equation (1)] for the composition $(d, d, \dots, d)$. The support polynomial $H_{n,d}$ must not be confused with the Ehrhart h^* -polynomial of $Q_{n,d}$ or with its ordinary face-enumeration h -polynomial. The last inequality makes the sum finite; the constant coefficient is 1, and the degree is dn with leading coefficient 1.

Put

$$B_N(t) = \sum_{j=0}^N \binom{N}{j}^2 t^j, \quad F_d(z, t) = \sum_{n \geq 0} H_{n,d}(t) z^n. \quad (1.3)$$

Specifying B_N by this formula removes any ambiguity in the indexing of Narayana polynomials: B_N is the type- $\mathbb{B}$ Narayana polynomial in the normalization used by the source. In a variable q , the source's displayed identity reads

$$1 + \sum_{n \geq 1} H_{n,d}(q^2) q^{-dn} x^n = \exp \left(\sum_{m \geq 1} B_{dm}(q^2) q^{-dm} \frac{x^m}{m} \right). \quad (1.4)$$

The substitution $x = zq^d$ removes all negative powers. Since $t \mapsto q^2$ is injective on polynomial coefficients, the target is equivalently

$$\boxed{F_d(z, t) = \exp \left(\sum_{m \geq 1} B_{dm}(t) \frac{z^m}{m} \right)}. \quad (1.5)$$

Removing the normalization is not cosmetic: it makes the specialization $t = 0$ legitimate and exposes an ordinary generating function in the number of equivalence classes. No parity restriction on d is needed.

1.4 What is proved, and what is not claimed

Section 2 proves (1.5) as a specialization of a rectangular theorem, and Section 3 proves the same rectangular theorem a second time by a different route. Sections 4–11 develop its consequences. Gamma positivity for *all* composition polytopes was already proved by Athanasiadis [2, Theorem 1.2]; recovering it for constant block sizes is not a new resolution of that question, and the two-subset encoding below is closely related to that paper's encoding. The formal factorization belongs to the classical Spitzer/Wiener–Hopf framework [3, 4]; it is not presented as a new general lemma, and the elementary form needed here is proved in full so that no inaccessible theorem is a hidden prerequisite. The contribution of the proofs is to identify the exact walks to which that factorization applies and thereby establish the selected identity.

We do not prove real-rootedness of $H_{n,d}$, gamma positivity for arbitrary preorders, flag realizability, or the weighted q -reciprocity conjecture in the same source. We do not claim that the algebraic degree bound 2^d is sharp, that $H_{n,d}$ is an Ehrhart h^* -polynomial, that the support

polynomial of an unequal-size block sequence depends only on the multiset of sizes (an explicit counterexample is given in Section 9), or that the asymptotics of Section 10 are uniform as d grows with n or as $t \rightarrow 0$. Those are different assertions. In particular, a generating-function identity with positive coefficients is not by itself a real-rootedness argument.

1.5 Deliberate redundancies

Because this article merges two independent treatments, several results are deliberately given twice. Each duplication is kept because the two versions are not substitutes; the list below states what each one buys.

1. **Two proofs of the headline identity** (Sections 2 and 3). The excursion proof gives a two-sided bounded symbol and a constant term evaluated by the binomial theorem; the slack-meander proof stays in the original coordinates, needs no encoding, and is the one that extends to unequal block sizes.
2. **Two forms of the factorization lemma** (Lemma 2.4 and Corollary 2.5). The Laurent-series form over $R((u^{-1}))$ is what the slack walk and the mixed-size theorem need; the constant-term excursion form is what the subset walk needs, and neither statement is a formal consequence of the other as stated.
3. **Two generalization directions** (Theorem 2.1 and Theorem 9.1). The rectangular budget cr is what powers the duality of Corollary 4.1 and the Ehrhart evaluation of (4.8); the commutative mixed-size formula is what covers unequal blocks, and neither implies the other.
4. **Two derivations of the one-block Narayana series** (Section 5.1). The first-step Motzkin decomposition is combinatorial and self-contained; the analytic route through $\sum_m B_m(t)w^m = \Delta^{-1/2}$ is what unlocks the Lagrange-inversion row (5.5).
5. **Two expressions for the asymptotic boundary constant** ((10.1) and (10.2)). The finite radical product is the fast evaluation; the convergent exponential sum is what identifies the constant as the boundary value $F_d(\rho_d(t), t)$.
6. **Two treatments of the gamma coefficients** (Sections 6.1 and 6.2). The block-boundary path model supplies a combinatorial meaning, a table, and the dictionary to the encoding of [2]; the algebraic constant-term derivation supplies integrality and strict positivity, which the path model does not give.
7. **Two degree-bound arguments for 2^d** (Theorem 5.1 and Proposition 5.2). The conjugate count is short; the explicit monic polynomial is constructive, and the two guard each other.
8. **Two elimination certificates for the double-chain quartic** (Corollary 5.3 and the second pair displayed after it). They eliminate different auxiliary quantities and are checked by two different scripts.
9. **Two independent verifier suites** (Section 12.1), over disjoint parameter ranges and with non-overlapping methods: only the first witnesses the subset encoding and the rectangular cases $c \neq d$, and only the second witnesses the endpoint factorization, the mixed-size recursion, and the range $d \leq 8, n \leq 12$.
10. **Two data corpora** (Appendix B). Neither is a subset of the other: one holds all 315 rectangular polynomials and the only κ_d table, the other reaches $d = 8$ and $n = 12$ and holds the only exact-versus-asymptotic ratio table.

11. **Two provenance records** (`sources.md`). One carries the hash and version reasoning for the pinned source, the other carries two adjacent 2026 papers and the Section 9.4 cross-check.

One duplication has deliberately *not* been kept: the weaker statement $\text{Var} = dn/8 + O(1)$ is strictly subsumed by the exact variance of Proposition 11.1 together with Theorem 11.2, so only the sharper form is stated. Both justifications of the quasi-power step are retained, however, in the proof of Theorem 11.2.

2 A rectangular lattice-point theorem, proved by excursions

2.1 Statement

For integers $d \geq 1$, $c \geq 0$, and $n \geq 0$, let

$$\mathcal{A}_{n;d,c} = \left\{ a \in \mathbb{N}^{dn} : \sum_{i=1}^r \sum_{j=1}^d a_{ij} \leq cr \quad (1 \leq r \leq n) \right\}, \quad A_{n;d,c}(t) = \sum_{a \in \mathcal{A}_{n;d,c}} t^{\#\text{supp}(a)}. \quad (2.1)$$

The notation $A_{0;d,c} = 1$ includes the empty array.

Theorem 2.1 (Rectangular exponential formula). *For every $d \geq 1$ and $c \geq 0$, the following identity holds in $\mathbb{Q}[t][[z]]$:*

$$\sum_{n \geq 0} A_{n;d,c}(t) z^n = \exp \left(\sum_{m \geq 1} \frac{z^m}{m} \sum_{j=0}^{\min(dm, cm)} \binom{dm}{j} \binom{cm}{j} t^j \right). \quad (2.2)$$

Consequently (1.5) holds for every $d \geq 1$. Both sides in fact have coefficients in $\mathbb{Z}[t]$.

The binomial product counts a single unrestricted bridge of m blocks. The exponential is not an approximation and is not inferred from matching initial coefficients. The next two lemmas explain exactly why it is the series for all excursions. Section 3 proves the same theorem again by a different walk.

2.2 From an array to two subsets

Flatten an array into $a_1, \dots, a_{dn}$. Suppose its positive coordinates occur at

$$S = \{s_1 < \dots < s_j\} \subseteq [dn].$$

Their successive positive partial sums are strictly increasing:

$$b_\ell = a_{s_1} + \dots + a_{s_\ell}, \quad T = \{b_1 < \dots < b_j\} \subseteq [cn]. \quad (2.3)$$

Here $[0]$ is the empty set. The case $j = 0$ gives $S = T = \emptyset$.

Lemma 2.2 (Subset encoding). *The map (2.3) is a support-preserving bijection between $\mathcal{A}_{n;d,c}$ and pairs*

$$S \subseteq [dn], \quad T \subseteq [cn], \quad |S| = |T|, \quad |S \cap [dr]| \leq |T \cap [cr]| \quad (1 \leq r \leq n). \quad (2.4)$$

The support size is $|S| = |T|$.

Proof. For the inverse, order S and T as above, put $b_0 = 0$, set $a_{s_\ell} = b_\ell - b_{\ell-1}$, and put all other coordinates equal to zero. This gives a unique nonnegative array with the specified support and partial sums. If $k = |S \cap [dr]|$, its sum over the first dr coordinates is b_k when $k > 0$, and zero when $k = 0$. The condition $b_k \leq cr$ is equivalent to having at least k elements of T in $[cr]$. This is precisely the last inequality in (2.4). The two constructions are inverse. $\square$

Divide the positions of S into n blocks of length d , and the positions of T into n blocks of length c . Write p_r and q_r for the numbers selected in their respective r th blocks. The successive walk heights are

$$h_r = \sum_{i=1}^r (q_i - p_i), \quad h_0 = h_n = 0, \quad h_r \geq 0. \quad (2.5)$$

A block with $(p_r, q_r) = (p, q)$ has $\binom{d}{p} \binom{c}{q}$ possible decorations, contributes weight t^p , and changes the height by $q - p$. Its Laurent-polynomial symbol is therefore

$$P_{d,c}(u, t) = \sum_{p=0}^d \sum_{q=0}^c \binom{d}{p} \binom{c}{q} t^p u^{q-p} = (1+u)^c (1+t/u)^d. \quad (2.6)$$

Thus $A_{n;d,c}(t)$ is the weight of all length- n nonnegative excursions with step symbol $P_{d,c}$. Positivity is checked after a whole block; no extra restriction is imposed on an ordering of choices inside it.

Example 2.3. Take $d = c = 2$, $n = 2$, and the array $(0, 2 \mid 1, 0)$. Then $S = \{2, 3\}$ and $T = \{2, 3\}$. Both block heights are zero and the weight is t^2 . For $(0, 0 \mid 0, 4)$, the sets are $S = \{4\}$ and $T = \{4\}$; the weight is t . The original coordinate partial sums and the subset-walk heights are different statistics. Only the specified equivalence of their prefix constraints is used.

2.3 The formal factorization, in two forms

Let R be any commutative algebra over $\mathbb{Q}$. We work with $R((u^{-1}))$, the ring of formal Laurent series in u whose exponents are bounded above; its products are well defined, since for any fixed exponent only finitely many pairs of exponents below the respective upper bounds contribute. For such a series, $[\cdot]_{\geq 0}$ retains its nonnegative powers of u and $[\cdot]_{< 0}$ retains the strictly negative powers. Apply these operators coefficientwise to series in z . At a fixed z -degree only finitely many factors contribute to a product, logarithm, or exponential with positive z -valuation, so all formal logarithms, exponentials, products, and projections used below are legitimate.

The first form is stated in this full generality, with infinitely many possible negative jumps and no positivity assumption on their weights. It is what Section 3 and Theorem 9.1 need.

Lemma 2.4 (Nonnegative-walk factorization). *Let $S(u) \in R((u^{-1}))$. There is a unique $M(z, u) \in R[[u]][[z]]$ with*

$$M = 1 + z[SM]_{\geq 0}, \quad (2.7)$$

namely

$$M(z, u) = \exp \left(\sum_{m \geq 1} \frac{z^m}{m} [S(u)^m]_{\geq 0} \right). \quad (2.8)$$

Proof. Set

$$L = -\log(1 - zS) = \sum_{m \geq 1} \frac{z^m S^m}{m}, \quad L_+ = [L]_{\geq 0}, \quad L_- = [L]_{< 0},$$

the projections acting coefficientwise in z . Each coefficient of L_+ is a polynomial in u , so $M_* := \exp(L_+)$ lies in $R[u][[z]]$. Since the coefficient algebra is commutative,

$$(1 - zS)M_* = \exp(-L) \exp(L_+) = \exp(-L_-). \quad (2.9)$$

Every monomial of L_- has strictly negative u -exponent, hence so does every nonconstant term of $\exp(-L_-)$; therefore $[\exp(-L_-)]_{\geq 0} = 1$. Applying $[\cdot]_{\geq 0}$ to (2.9), and using that M_* has only nonnegative powers of u , gives $M_* - z[SM_*]_{\geq 0} = 1$, so M_* solves (2.7).

Finally, writing $M = \sum_{n \geq 0} M_n(u)z^n$, equation (2.7) determines $M_0 = 1$ and $M_n = [SM_{n-1}]_{\geq 0}$ for $n \geq 1$. This proves existence and uniqueness, and hence (2.8). $\square$

Corollary 2.5 (Excursions are the exponential of bridges). *Let $P(u) \in R[u, u^{-1}]$ be a Laurent polynomial and let $E_P(z)$ be the generating series of walks with step symbol P which start and end at height zero and remain at nonnegative heights. Then*

$$E_P(z) = \exp \left(\sum_{m \geq 1} \frac{z^m}{m} \text{CT}_u P(u)^m \right). \quad (2.10)$$

Proof. Because P has a finite exponent range, each coefficient of the meander series M_P of Lemma 2.4 is a Laurent polynomial, and $E_P = \text{CT}_u M_P$. Now L_+ has only nonnegative u -degrees, so a product of its monomials can have degree zero only when every chosen monomial has degree zero; hence

$$\text{CT}_u \exp(L_+) = \exp(\text{CT}_u L_+) = \exp(\text{CT}_u L),$$

which is (2.10). $\square$

Remark 2.6. Neither statement contains the other as written. Lemma 2.4 is needed in $R((u^{-1}))$, where the slack symbol (3.2) lives and where the mixed-size step series of Section 9 lives; Corollary 2.5 needs a two-sided bounded symbol for E_P to be the excursion enumerator with finitely many decorated steps per block. The constant-term algebra in the proof of the corollary does extend verbatim to $R((u^{-1}))$; what does not extend is the reading of $\text{CT}_u P^m$ as a finite bridge count over a bounded step set, which is exactly what the binomial theorem evaluates in Section 2.4.

The separation of degree zero into the *nonnegative* factor is essential: the walks may touch zero at intermediate block boundaries. Replacing this by a strictly positive convention would count a different class. One must likewise not replace the proof by the assertion that every cyclic orbit has exactly one nonnegative rotation: zero sums, periodic words, and unequal positive jumps make that assertion false in this form. The factorization handles these degeneracies without any orbit-counting assumption.

2.4 First proof of the theorem

First proof of Theorem 2.1. Lemma 2.2 and (2.6) identify the left side of (2.2) with $E_{P_{d,c}}(z)$. The binomial theorem gives

$$\begin{aligned} \text{CT}_u P_{d,c}(u, t)^m &= \text{CT}_u (1 + u)^{cm} (1 + t/u)^{dm} \\ &= \sum_{j=0}^{\min(dm, cm)} \binom{cm}{j} \binom{dm}{j} t^j. \end{aligned}$$

Substitution in Corollary 2.5 proves (2.2). For $c = d$, the defining arrays are exactly the lattice points of (1.1), so $A_{n;d,c} = H_{n,d}$ and the selected conjecture follows. If $c = 0$, only the zero array is possible for each n ; both sides reduce to $(1 - z)^{-1}$, consistently with the same proof. The lattice-point interpretation places all coefficients in $\mathbb{Z}[t]$. $\square$

3 A second proof by the original slack process

This section proves Theorem 2.1 a second time, without the subset encoding, by running the process on the original coordinates. The walk, the symbol, the ring and the boundary condition are all different from those of Section 2: the admissible arrays are meanders with a free terminal slack rather than excursions, the symbol is a Laurent series with an infinite negative tail rather than a Laurent polynomial, and the logarithm is extracted by the nonnegative projection followed by $u = 1$ rather than by a constant term. The argument also produces an endpoint refinement that the first proof does not see.

3.1 Slack walks and the block weight

For an array in (2.1), let $Y_r = \sum_{j=1}^d a_{rj}$ and set

$$k_r = cr - \sum_{i=1}^r Y_i, \quad k_0 = 0.$$

The inequalities say exactly that $k_r \geq 0$ for every r ; the final slack k_n may be arbitrary. The jump in slack is $c - Y_r$, bounded above by c , but unbounded below before nonnegativity is imposed. Thus the arrays are *meanders*, not excursions, of this walk.

Retaining the whole contents of a block as its decoration, a block of d coordinates with prescribed coordinate sum s has support polynomial

$$W_{d,s}(t) = \begin{cases} 1, & s = 0, \\ \sum_{r=1}^{\min(d,s)} \binom{d}{r} \binom{s-1}{r-1} t^r, & s \geq 1, \end{cases} \quad (3.1)$$

because the second binomial coefficient counts positive compositions of s into r specified coordinates. Formula (3.1) yields a finite-state dynamic program on the slack, using only the defining inequalities and no conjectured identity; this is what one of the two verifier suites of Section 12.1 actually runs.

3.2 The step symbol and its ring

The weight generating series of one coordinate is

$$\omega(v, t) = 1 + \frac{tv}{1-v} = \sum_{s \geq 0} W_{1,s}(t) v^s.$$

The slack-step symbol, expanded at $u = \infty$, is

$$S_{d,c}(u, t) = u^c \omega(u^{-1}, t)^d = u^c \left(\frac{u+t-1}{u-1} \right)^d = \sum_{s \geq 0} W_{d,s}(t) u^{c-s}. \quad (3.2)$$

Here and below $1/(u-1) = u^{-1} + u^{-2} + u^{-3} + \dots$ is expanded at infinity; expansion at $u = 0$ would encode a different walk and invalidate the argument. The symbol has largest u -degree c and an infinite negative tail, so it lies in $\mathbb{Q}[t]((u^{-1}))$ and not in $\mathbb{Q}[t][u, u^{-1}]$. Lemma 2.4 applies verbatim with $R = \mathbb{Q}[t]$: each z^m coefficient is bounded above in u -degree, and its nonnegative projection is a finite polynomial.

3.3 The endpoint refinement

Let

$$M_{d,c}(z, u, t) = \sum_{n \geq 0} z^n \sum_{a \in \mathcal{A}_{n;d,c}} t^{\#\text{supp}(a)} u^{cn - \sum_{i,j} a_{ij}}$$

record the terminal slack as well as the support. Appending one block and discarding negative new slack is exactly (2.7) for $S = S_{d,c}$, so Lemma 2.4 gives

$$M_{d,c}(z, u, t) = \exp \left(\sum_{m \geq 1} \frac{z^m}{m} [S_{d,c}(u, t)^m]_{\geq 0} \right). \quad (3.3)$$

Corollary 3.1 (Endpoint-refined factorization). *Since $S_{d,c}(u, t)^m = S_{dm,cm}(u, t)$, one has for $cm > 0$*

$$[S_{d,c}(u, t)^m]_{\geq 0} = \sum_{s=0}^{cm} W_{dm,s}(t) u^{cm-s} = u^{cm} + \sum_{j=1}^{\min(dm,cm)} \binom{dm}{j} t^j \sum_{\ell=j}^{cm} \binom{\ell-1}{j-1} u^{cm-\ell}, \quad (3.4)$$

and therefore

$$M_{d,c}(z, u, t) = \exp \left(\sum_{m \geq 1} \frac{z^m}{m} \sum_{s=0}^{cm} W_{dm,s}(t) u^{cm-s} \right). \quad (3.5)$$

When $c = 0$ the projection is simply 1.

Proof. Both displays in (3.4) are the truncation of (3.2) with (d, c) replaced by (dm, cm) to the range $s \leq cm$; expanding $W_{dm,s}$ by (3.1) and exchanging the two summations gives the second form with $\ell = s$. $\square$

This refinement computes the joint distribution of support and unused budget. Its diagonal case $c = d$ is verified directly at integer t by one of the verifier suites of Section 12.1.

3.4 Evaluating the positive part

Lemma 3.2 (Unrestricted endpoint count; stars and bars). *For all integers $D \geq 0$ and $C \geq 0$,*

$$\sum_{\substack{a_1, \dots, a_D \geq 0 \\ a_1 + \dots + a_D \leq C}} t^{\#\{i: a_i > 0\}} = \sum_{r \geq 0} \binom{D}{r} \binom{C}{r} t^r. \quad (3.6)$$

Proof. Choose the r positive coordinates in $\binom{D}{r}$ ways. Subtract one from each of them; introducing a nonnegative slack variable converts their sum-at-most- C condition into an equation with $r + 1$ nonnegative unknowns and total $C - r$, which stars and bars counts in $\binom{C}{r}$ ways. The same formula at $r = 0$ counts the zero vector. Multiplying the two choices and summing proves the identity. $\square$

Second proof of Theorem 2.1. Equation (3.3) holds by Lemma 2.4. Each coefficient of that series is a polynomial in u , so the substitution $u = 1$ is a well-defined ring homomorphism applied coefficientwise in z . The projected polynomial at index m , evaluated at $u = 1$, enumerates all dm coordinates with total at most cm ; Lemma 3.2 with $D = dm$ and $C = cm$ turns this into $\sum_j \binom{dm}{j} \binom{cm}{j} t^j$. Since $F_{d,c}(z, t) = M_{d,c}(z, 1, t)$, this is (2.2) again. $\square$

Substituting $u = 1$ directly into the rational expression (3.2) would be invalid, because it has a pole there. Projecting first is what makes the proof correct. Both proofs are independent of the manifest's reciprocity, root-polytope, and triangulation claims: each uses only the defining prefix inequalities, a formal factorization, and an elementary count.

3.5 Why the type- $\mathbb{B}$ polynomial appears

The coefficient in the logarithm does *not* impose the intermediate constraints at block boundaries. It imposes only nonnegative *final* slack after m blocks. Those dm coordinates then satisfy a single simplex constraint, and the two binomial choices in Lemma 3.2 – positions and values – explain the product $\binom{dm}{j} \binom{cm}{j}$, and hence the square when $c = d$. The exponential restores all intermediate nonnegativity conditions through the factorization. This separation is the structural content of the identity.

4 Duality, recurrences, and exact enumeration

4.1 A support-preserving rectangular duality

Corollary 4.1. *For $c, d \geq 1$ and all n , one has*

$$A_{n;d,c}(t) = A_{n;c,d}(t). \quad (4.1)$$

There is an explicit support-preserving bijection realizing this equality.

Proof. The generating-function proof follows from the symmetry of the binomial product. For a bijection, encode an array by (S, T) and put

$$S' = \{cn + 1 - b : b \in T\}, \quad T' = \{dn + 1 - s : s \in S\}.$$

If h_r is the original block height, the new height at block r is h_{n-r} : swap the two subset types and reverse the block sequence. It is nonnegative, and the new path also ends at zero. Decode (S', T') using Lemma 2.2 with the dimensions interchanged. Applying this map twice restores the original pair, and its common cardinality is unchanged. $\square$

For example, $A_{1;2,3}(t) = 1 + 6t + 3t^2 = A_{1;3,2}(t)$. The arrays live in different ambient dimensions, so this equality is not coordinate reversal of the original array. The subset description is what makes the bijection simple; this corollary has no counterpart in the diagonal case alone.

4.2 Convolution and partition formulas

Define the bridge polynomial

$$C_m^{d,c}(t) = \sum_j \binom{dm}{j} \binom{cm}{j} t^j. \quad (4.2)$$

Logarithmic differentiation of (2.2) immediately gives

$$\boxed{nA_{n;d,c}(t) = \sum_{m=1}^n C_m^{d,c}(t) A_{n-m;d,c}(t), \quad A_{0;d,c}(t) = 1.} \quad (4.3)$$

Although this recursion divides by n , Theorem 2.1 proves that the division is exact coefficientwise over $\mathbb{Z}[t]$. The accompanying implementations check this divisibility rather than rounding numerical values.

For a partition $\lambda \vdash n$, let $m_r(\lambda)$ be its number of parts of size r and put $z_\lambda = \prod_{r \geq 1} r^{m_r} m_r!$. Expanding the exponential also gives the finite formula

$$A_{n;d,c}(t) = \sum_{\lambda \vdash n} \frac{1}{z_\lambda} \prod_{r \geq 1} C_r^{d,c}(t)^{m_r(\lambda)}. \quad (4.4)$$

Equivalently, because the number of permutations of cycle type λ is $n!/z_\lambda$,

$$A_{n;d,c}(t) = \frac{1}{n!} \sum_{\pi \in \mathfrak{S}_n} \prod_{\gamma \text{ a cycle of } \pi} C_{|\gamma|}^{d,c}(t). \quad (4.5)$$

This is a cycle-index identity, or weighted permutation average; it is not, without further work, a bijection between lattice points and unlabelled collections of decorated cycles.

In the diagonal case the first rows can be recorded symbolically in d :

$$\begin{aligned} H_{2,d} &= \frac{B_d^2 + B_{2d}}{2}, & H_{3,d} &= \frac{B_d^3 + 3B_d B_{2d} + 2B_{3d}}{6}, \\ H_{4,d} &= \frac{B_d^4 + 6B_d^2 B_{2d} + 3B_{2d}^2 + 8B_d B_{3d} + 6B_{4d}}{24}. \end{aligned} \quad (4.6)$$

At $t = 1$, Vandermonde's identity yields

$$C_m^{d,c}(1) = \binom{(c+d)m}{dm}, \quad B_m(1) = \binom{2m}{m}. \quad (4.7)$$

Thus the totals can be evaluated through index N using $O(N^2)$ integer arithmetic operations after precomputing the binomial coefficients. With naive coefficient convolution, computing all support polynomials through index N takes $O(b^2 N^4)$ arithmetic operations and $O(bN^2)$ stored coefficients, where $b = \min(c, d)$ is fixed and positive. These are arithmetic operation counts, not bit-complexity bounds.

4.3 Examples and boundary cases

For one block the prefix condition is just a total-sum bound, so $H_{1,d} = B_d$. The first double-chain polynomials are

$$\begin{aligned} H_{1,2}(t) &= 1 + 4t + t^2, \\ H_{2,2}(t) &= 1 + 12t + 27t^2 + 12t^3 + t^4, \\ H_{3,2}(t) &= 1 + 24t + 134t^2 + 236t^3 + 134t^4 + 24t^5 + t^6. \end{aligned}$$

For instance, $2H_{2,2} = B_2^2 + B_4$ follows directly from the recurrence, and $H_{2,2}(1) = 53$. These values agree with the small double-chain table in [1, Section 9.4]. Table 1 gives further totals computed independently from coordinate dynamic programming and from (4.3).

n	$H_{n,1}(1)$	$H_{n,2}(1)$	$H_{n,3}(1)$	$H_{n,4}(1)$
0	1	1	1	1
1	2	6	20	70
2	5	53	662	8885
3	14	554	26780	1409002
4	42	6362	1205961	250837850
5	132	77580	58050204	47928217484
6	429	986253	2924165436	9605146588557
7	1430	12927170	152231599628	1992173405162370

Table 1: Exact lattice-point counts. The $d = 1$ column is the Catalan sequence shifted by one index.

The specialization $t = 0$ gives $H_{n,d}(0) = 1$ and $F_d(z, 0) = (1 - z)^{-1}$, as it must, since only the zero lattice point remains. The leading coefficient of $H_{n,d}$ is also 1: if all dn coordinates are positive, the total bound dn forces every coordinate to be 1. These are useful normalization checks; changing the source's support statistic or number of blocks would usually violate one of them.

4.4 Uniform dilations and Ehrhart evaluation

For a nonnegative integer ℓ , the lattice points of $\ell Q_{n,d}$ are precisely $\mathcal{A}_{n;d,\ell,d}$. Therefore Theorem 2.1 gives

$$\sum_{n \geq 0} \text{Ehr}(Q_{n,d}, \ell) z^n = \exp \left(\sum_{m \geq 1} \binom{(\ell+1)dm}{dm} \frac{z^m}{m} \right). \quad (4.8)$$

The corresponding finite partition formula is

$$\text{Ehr}(Q_{n,d}, \ell) = \sum_{\lambda \vdash n} \frac{1}{z_\lambda} \prod_{r \geq 1} \binom{(\ell+1)dr}{dr}^{m_r(\lambda)}. \quad (4.9)$$

For fixed n, d , its right side is a polynomial in ℓ of degree at most dn . Thus this is also an identity of Ehrhart polynomials, since it holds at every nonnegative integer ℓ . It provides an evaluation formula in the number of blocks; it is not a proof of real-rootedness of those polynomials, and it does not identify the support polynomial $H_{n,d}$ with an Ehrhart h^* -polynomial. This evaluation is available only because Theorem 2.1 is rectangular.

5 Algebraic generating functions

5.1 The one-block-size series, two ways

For $d = c = 1$, the subset-walk symbol is

$$P_{1,1}(u, t) = 1 + t + u + t/u. \quad (5.1)$$

These are Motzkin excursions with horizontal weight $1 + t$, up weight 1, and down weight t . A nonempty excursion begins either with a horizontal step followed by an excursion, or with an up step, an excursion translated up one unit, a down step, and another excursion. Hence, writing $g(w, t) = F_1(w, t)$,

$$g = 1 + (1 + t)wg + tw^2g^2. \quad (5.2)$$

The unique formal solution with constant term 1 is

$$\boxed{g(w, t) = \frac{2}{1 - (1 + t)w + \sqrt{\Delta(w, t)}}}, \quad \Delta(w, t) = 1 - 2(1 + t)w + (1 - t)^2w^2. \quad (5.3)$$

The square root has constant term 1. This rationalized expression remains valid at $t = 0$, unlike a formula which divides by t . At $t = 1$ it becomes $g(w, 1) = \text{Cat}(w)^2$, where $\text{Cat}(w) = 2/(1 + \sqrt{1 - 4w})$ is the Catalan generating function, so $H_{n,1}(1)$ is the Catalan number Cat_{n+1} .

A second, independent derivation runs through the ordinary generating function of the Narayana polynomials themselves. Summing the constant-term expression (6.6) below gives

$$\sum_{m \geq 0} B_m(t)w^m = \frac{1}{\sqrt{\Delta(w, t)}}, \quad (5.4)$$

the square root again having constant coefficient 1. Indeed, expand $(1 - (1 + t)w - \sqrt{t}w(v + v^{-1}))^{-1}$ in powers of $v + v^{-1}$ and take its constant term in v : only even powers survive, with constant term $\binom{2r}{r}$, and summing them gives (5.4). Now Theorem 2.1 at $d = 1$ says $\log g = \sum_{m \geq 1} B_m(t)w^m/m$, so

$$w \frac{\partial}{\partial w} \log g = \frac{1}{\sqrt{\Delta(w, t)}} - 1,$$

whose unique solution with constant coefficient 1 is again (5.3).

The second route is what yields a closed form for the $d = 1$ row. Put $Y = wg$; then (5.2) becomes $Y = w(1 + Y)(1 + tY)$, and Lagrange inversion gives the Narayana row

$$H_{n,1}(t) = \frac{1}{n+1} \sum_{j=0}^n \binom{n+1}{j} \binom{n+1}{j+1} t^j. \quad (5.5)$$

5.2 A root-of-unity product for every block size

Theorem 5.1. *Let $d \geq 1$, let $\xi = e^{2\pi i/d}$, and introduce a formal variable s with $s^d = z$. Then*

$$F_d(z, t) = \prod_{r=0}^{d-1} g(\xi^r s, t). \quad (5.6)$$

The product contains only powers of s^d , has rational polynomial coefficients in t , and is algebraic over $\mathbb{Q}(z, t)$ of degree at most 2^d . No assertion that this upper bound is always minimal is made.

Proof. Theorem 2.1 with $d = 1$ gives $\log g(w, t) = \sum_{m \geq 1} B_m(t) w^m / m$. Therefore

$$\begin{aligned} \log \prod_{r=0}^{d-1} g(\xi^r s, t) &= \sum_{m \geq 1} \frac{B_m(t) s^m}{m} \sum_{r=0}^{d-1} \xi^{rm} \\ &= \sum_{k \geq 1} \frac{B_{dk}(t) s^{dk}}{k} = \log F_d(s^d, t). \end{aligned}$$

Both series have constant coefficient 1, so the formal exponential proves the identity and the cancellation of fractional powers. Algebraicity follows already from (5.3), since each factor is quadratic over $\mathbb{Q}(t, s)$ and algebraicity is transitive.

For the degree bound, work over the rational-function field $\mathbb{Q}(z, t)$ and adjoin s , all d th roots of unity, and the d square roots occurring in the factors of (5.6). This is contained in a finite normal extension. An automorphism over $\mathbb{Q}(z, t)$ permutes the arguments $\xi^r s$, and can change the signs of their square roots. As the product is insensitive to the permutation, every conjugate is one of at most 2^d sign-choice products. In characteristic zero the algebraic degree is the number of distinct conjugates, proving the bound. $\square$

Proposition 5.2 (A constructive form of the same bound). *Let g_r^+ and g_r^- be the two roots of the quadratic (5.2) at $w = \xi^r s$, in a splitting field over $\mathbb{Q}(z, t)$ containing all d roots $\xi^r s$ of $s^d = z$. Then*

$$\prod_{\varepsilon \in \{+, -\}^d} \left(T - \prod_{r=0}^{d-1} g_r^{\varepsilon_r} \right) \quad (5.7)$$

is a monic polynomial of degree 2^d with coefficients in $\mathbb{Q}(z, t)$, and $F_d(z, t)$ is one of its roots.

Proof. Every automorphism of the splitting field over $\mathbb{Q}(z, t)$ permutes the $\xi^r s$ and, over each of them, either exchanges or preserves the two quadratic roots. It therefore permutes the 2^d displayed factors, so the coefficients of (5.7) are fixed and lie in $\mathbb{Q}(z, t)$. The branch in (5.6) is the factor with all $\varepsilon_r = +$. $\square$

The conjugate count of Theorem 5.1 is shorter; the explicit polynomial (5.7) is constructive and exhibits the annihilating equation. Keeping both is cheap and they guard each other.

This product is a multiplicative root-of-unity filter. An *average* of the $g(\xi^r s, t)$ would instead extract the coefficients $H_{dn,1}$; those count paths constrained after every microstep and generally give the wrong answer. For example, $H_{1,2}(1) = 6$ whereas $H_{2,1}(1) = 5$.

Algebraicity also implies that F_d satisfies a linear differential equation over $\mathbb{Q}(z, t)$, and its coefficients in z satisfy a polynomial-coefficient recurrence in n . Indeed, derivatives of an algebraic function lie in the same finite extension of $\mathbb{Q}(z, t)$, so sufficiently many are linearly dependent over that field. This observation does not turn (4.3) itself into a fixed-order, constant-coefficient recurrence.

5.3 An explicit quartic for double chains

Corollary 5.3. *The double-chain series $Y = F_2(z, t)$ is the unique solution in $\mathbb{Q}[t][[z]]$ with constant term 1 of*

$$\begin{aligned} 0 = & t^4 z^4 Y^4 + t^2 z^2 ((1+t)^2 z - 1) Y^3 \\ & + 2tz(1 + (t^2 + t + 1)z) Y^2 + ((1+t)^2 z - 1) Y + 1. \end{aligned} \quad (5.8)$$

Proof. Write $z = s^2$, $a = g(s, t)$, $b = g(-s, t)$, and $Y = ab$. The two quadratic relations (5.2) give

$$\begin{aligned} ts^2 a^2 + ((1+t)s - 1)a + 1 &= 0, \\ ts^2 Y^2 + (-(1+t)s - 1)Ya + a^2 &= 0. \end{aligned}$$

Their resultant with respect to a , followed by $s^2 = z$, is exactly the right side of (5.8). This polynomial identity is reproduced by `code/derive_quartic.py`. The defining polynomial has value $1 - Y$ at $z = 0$ and derivative -1 with respect to Y there. Coefficientwise induction, or the formal implicit-function theorem, proves uniqueness of the branch with constant term 1. $\square$

A second elimination certificate, computed by a different script from a different pair, reaches the same quartic. With $a = g(w, t)$ and $F = a g(-w, t)$, the pair is

$$\begin{aligned} tw^2 a^2 + ((1+t)w - 1)a + 1 &= 0, \\ a^2 - ((1+t)w + 1)Fa + tw^2 F^2 &= 0, \end{aligned} \quad (5.9)$$

whose resultant in a , after $w^2 = z$, is again the right side of (5.8). The script `code/symbolic_certificate.py` verifies that this difference is the zero polynomial, rather than a series vanishing to finite order.

At $t = 1$, the equation is

$$z^4 Y^4 + (4z^3 - z^2) Y^3 + (6z^2 + 2z) Y^2 + (4z - 1) Y + 1 = 0.$$

The independent standard-library verifier also substitutes the exact support polynomials into (5.8) through z^{16} , retaining every power of t ; this is a check of the printed certificate, not a substitute for the resultant proof.

6 Gamma coefficients: a path model and an algebraic derivation

The gamma expansion itself is a single identity, stated once in (6.2) and (6.4). The two derivations below have different corollaries and are both retained.

6.1 The block-boundary path model

For $c = d$, one has

$$P_{d,d}(u, t) = (1 + t + u + t/u)^d. \quad (6.1)$$

Thus an excursion with n macrosteps is a word of dn microsteps, grouped into blocks of d . Up and down steps have increments $+1, -1$, and a horizontal step has increment zero and two colors of weights $1, t$. The walk must have nonnegative height only after the d th, $2d$ th, $\dots$, nd th microsteps and must finish at zero.

Proposition 6.1. *Let $\gamma_{n,d,j}$ be the number of uncolored words of length dn in $\{U, D, H\}$ having j up steps and j down steps, total height zero, and nonnegative heights at multiples of d . Then*

$$H_{n,d}(t) = \sum_{j=0}^{\lfloor dn/2 \rfloor} \gamma_{n,d,j} t^j (1+t)^{dn-2j}. \quad (6.2)$$

In particular, $H_{n,d}$ is palindromic and unimodal.

Proof. Expand each factor of (6.1) into its four choices. An excursion has equally many up and down steps. If that number is j , the vertical steps together contribute t^j , while the $dn - 2j$ horizontal positions contribute $(1+t)^{dn-2j}$. Summing over uncolored words proves the expansion. Each displayed summand has a symmetric unimodal coefficient sequence with center $dn/2$; their nonnegative sum has the same properties. $\square$

For $d = 2$, the first gamma vectors are

n	$(\gamma_{n,2,0}, \gamma_{n,2,1}, \dots)$
0	(1)
1	(1, 2)
2	(1, 8, 5)
3	(1, 18, 47, 14)
4	(1, 32, 190, 244, 42)
5	(1, 50, 530, 1560, 1186, 132)

For $n = 1, d = 2$, both UD and DU are allowed, explaining the gamma coefficient 2. The word DU would be forbidden for ordinary Motzkin paths, which require nonnegativity after every microstep.

This is a constant-block-size reformulation of the earlier composition polytope gamma phenomenon [2, Theorem 1.2 and Proposition 2.2]. In the two-subset encoding, the common elements of S and T give the horizontal color of weight t , the elements in neither give the other color, and $T \setminus S$ and $S \setminus T$ give up and down steps. Requiring $S \cap T = \emptyset$ leaves exactly the uncolored paths counted by $\gamma_{n,d,j}$. This makes the connection to the prior interpretation explicit rather than treating positivity as an uncredited new theorem.

If $\Gamma_{n,d}(y) = \sum_j \gamma_{n,d,j} y^j$, so that

$$H_{n,d}(t) = (1+t)^{dn} \Gamma_{n,d}\left(\frac{t}{(1+t)^2}\right), \quad (6.3)$$

the same factorization gives

$$\sum_{n \geq 0} \Gamma_{n,d}(y) z^n = \exp\left(\sum_{m \geq 1} \frac{z^m}{m} G_{dm}(y)\right), \quad G_N(y) = \sum_{r=0}^{\lfloor N/2 \rfloor} \binom{N}{2r} \binom{2r}{r} y^r. \quad (6.4)$$

Indeed, use the uncolored symbol $(1 + u + y/u)^d$; a bridge of dm microsteps chooses its $2r$ vertical positions and then its r up positions.

6.2 The algebraic derivation, integrality and strict positivity

The same G_N arises from a constant-term computation, which yields properties the path model does not. Note first that

$$\binom{N}{2r} \binom{2r}{r} = \frac{N!}{r!^2 (N-2r)!}, \quad (6.5)$$

so that, with v an auxiliary variable and $\sqrt{t}$ a formal symbol,

$$\begin{aligned} B_N(t) &= \text{CT}_v (1 + \sqrt{t}v)^N (1 + \sqrt{t}v^{-1})^N \\ &= \sum_{r=0}^{\lfloor N/2 \rfloor} \frac{N!}{r!^2 (N-2r)!} t^r (1+t)^{N-2r} = (1+t)^N G_N \left(\frac{t}{(1+t)^2} \right). \end{aligned} \quad (6.6)$$

The displayed constant term is a polynomial in t ; the square root is only an auxiliary symbol. Applying (6.6) factorwise to (4.4) in the diagonal case gives (6.3) and (6.4) directly, with the total factor of $1+t$ equal to $(1+t)^{dn}$ because the parts of λ sum to n .

Proposition 6.2 (Integrality and strict positivity). *Every $\gamma_{n,d,j}$ is a nonnegative integer, $\gamma_{n,d,0} = 1$, and for $n \geq 1$ every $\gamma_{n,d,j}$ with $0 \leq j \leq \lfloor dn/2 \rfloor$ is strictly positive. Consequently the coefficients of $H_{n,d}$ increase strictly up to the centre, except for the equal middle pair forced by odd degree.*

Proof. The exponential (6.4) produces a priori only nonnegative rational gamma coefficients. For integrality, observe that the coefficients of any symmetric integer polynomial of degree N in the basis $t^r(1+t)^{N-2r}$ are integers: determine the coefficient at $r=0$, subtract that multiple, and continue successively with the coefficient of t^r . Each step subtracts an integer polynomial and requires no division. Applying this to $H_{n,d}$, whose coefficients are integers by Theorem 2.1, gives integrality.

For strict positivity, expand the exponential over partitions as in (4.4). All contributions are nonnegative, and the single-part partition (n) alone contributes $G_{dn}(y)/n$, whose coefficient of y^j is positive for every $0 \leq j \leq \lfloor dn/2 \rfloor$. At $y=0$ the exponential in (6.4) is $(1-z)^{-1}$, so $\gamma_{n,d,0} = 1$; the case $n=0$ is immediate. The final statement follows because the $r=0$ term of (6.3) is $(1+t)^{dn}$, whose binomial coefficients increase strictly to the centre, and all other terms are nonnegative and symmetric about the same centre. $\square$

Writing out the coefficient of y gives the closed form

$$\gamma_{n,d,1} = \frac{d^2 n(n+1)}{2} - nd, \quad (6.7)$$

which also follows from the first support coefficient (7.6) below by subtracting nd , the coefficient of t in the gamma-basis term $(1+t)^{dn}$. As recorded in Section 1, qualitative gamma positivity was already proved in greater generality in [2]; what (6.4) adds here is an explicit exponential organization in the equal-block case, and what Proposition 6.2 adds is integrality together with strictness.

7 Fixed support size: polynomiality in the number of blocks

Write $h_{n,d,j} = [t^j]H_{n,d}(t)$, with $h_{n,d,j} = 0$ if $j > dn$. The following result concerns a fixed j , not the whole row, whose degree grows with n . It has no counterpart in the rectangular development above.

Theorem 7.1 (Polynomiality and leading coefficient). *Fix integers $d \geq 1$ and $j \geq 1$. As a function of $n \in \mathbb{N}$, $h_{n,d,j}$ is the restriction of a rational-coefficient polynomial of exact degree $2j$, with leading coefficient*

$$\boxed{\frac{d^{2j}}{j!(j+1)!}}. \quad (7.1)$$

Equivalently, its ordinary generating function has the form

$$\sum_{n \geq 0} h_{n,d,j} z^n = \frac{\Pi_{d,j}(z)}{(1-z)^{2j+1}}, \quad \deg \Pi_{d,j} \leq 2j-1, \quad (7.2)$$

with

$$\Pi_{d,j}(1) = \text{Cat}_j d^{2j}, \quad \text{Cat}_j = \frac{1}{j+1} \binom{2j}{j}. \quad (7.3)$$

In particular the pole order $2j+1$ is exact.

Proof. Separate the constant-in- t part of the logarithm in (1.5):

$$F_d(z, t) = \frac{1}{1-z} \exp \left(\sum_{r \geq 1} t^r \Lambda_{d,r}(z) \right), \quad \Lambda_{d,r}(z) = \sum_{m \geq 1} \binom{dm}{r}^2 \frac{z^m}{m}. \quad (7.4)$$

For $r \geq 1$, the expression $\binom{dm}{r}^2/m$ is a polynomial in m of degree $2r-1$: indeed $\binom{dm}{r}$ has a factor m , so the quotient is polynomial and has zero constant term. The elementary identity

$$\sum_{m \geq 1} m^a z^m = \left(z \frac{d}{dz} \right)^a \frac{z}{1-z}$$

shows that $\Lambda_{d,r}$ has denominator dividing $(1-z)^{2r}$ and a numerator of degree at most $2r-1$.

In the coefficient of t^j in the exponential, a product $\prod_r \Lambda_{d,r}^{e_r}$ satisfies $\sum_r r e_r = j$; its denominator divides $(1-z)^{2j}$ and its numerator has degree at most $2j - \sum_r e_r \leq 2j-1$. The extra factor $(1-z)^{-1}$ in (7.4) proves (7.2).

For the highest pole, the leading term of $\binom{dm}{r}^2/m$ is $d^{2r} m^{2r-1}/(r!)^2$, so

$$\Lambda_{d,r}(z) = \frac{d^{2r}(2r-1)!}{(r!)^2} (1-z)^{-2r} + O((1-z)^{-2r+1}). \quad (7.5)$$

The coefficient of the highest pole in (7.2) is therefore $[\sigma^j]$ of

$$\exp \left(\sum_{r \geq 1} \frac{1}{2r} \binom{2r}{r} (d^2 \sigma)^r \right) = \text{Cat}(d^2 \sigma),$$

the last identity following, for instance, by differentiating $\log \text{Cat}(v)$ and using $\text{Cat}(v) = 2/(1 + \sqrt{1-4v})$. This proves (7.3).

Finally, each monomial z^a in $\Pi_{d,j}$ contributes $\binom{n+2j-a}{2j}$ to the coefficient of z^n . Since $a \leq 2j-1$, this binomial expression is a polynomial in n agreeing with the coefficient for every $n \geq 0$, including the initial zero values, and its leading coefficient is $1/(2j)!$. Hence the leading coefficient of $h_{n,d,j}$ is $\Pi_{d,j}(1)/(2j)! = d^{2j}/(j!(j+1)!)$. $\square$

7.1 The first two coefficients

At support size one, either (7.4) or a direct coordinate count gives

$$h_{n,d,1} = d^2 \binom{n+1}{2}. \quad (7.6)$$

For the direct argument, a coordinate in block b may contain any positive integer at most bd if all other coordinates vanish; there are d positions in that block, and $\sum_{b=1}^n d(bd) = d^2 \binom{n+1}{2}$.

At support size two,

$$\begin{aligned} h_{n,d,2} = & \frac{d^2(d-1)^2}{4} \binom{n+3}{4} + \frac{3d^4 - d^2}{2} \binom{n+2}{4} \\ & + \frac{d^2(d+1)^2}{4} \binom{n+1}{4}. \end{aligned} \quad (7.7)$$

To check this directly, use $[t^2]F_d = (1-z)^{-1}(\Lambda_{d,2} + \frac{1}{2}\Lambda_{d,1}^2)$; put it over $(1-z)^5$, whose numerator is

$$\frac{1}{4}(d^4(z + 6z^2 + z^3) - 2d^3(z - z^3) + d^2(z - 2z^2 + z^3)),$$

and extract coefficients. The leading coefficients of (7.6) and (7.7) are $d^2/2$ and $d^4/12$, in agreement with Theorem 7.1.

8 A Stieltjes moment representation in the number of blocks

Fix a real number $t > 0$. The variable in this section is the number of blocks n , not the support degree. Let U be the bilateral shift on $\ell^2(\mathbb{Z})$ and define

$$J = (1+t)I + \sqrt{t}(U + U^*) = (I + \sqrt{t}U)(I + \sqrt{t}U^*). \quad (8.1)$$

This is a bounded positive self-adjoint operator. If P is the orthogonal projection onto $\ell^2(\mathbb{N})$, let

$$T_d = PJ^dP|_{\ell^2(\mathbb{N})}. \quad (8.2)$$

The compression is applied *after* the d th power. In general, PJ^dP and $(PJP)^d$ are different: the latter would disallow negative intermediate microstep heights.

Theorem 8.1. *For every $d \geq 1$ and $t > 0$,*

$$H_{n,d}(t) = \langle e_0, T_d^n e_0 \rangle \quad (n \geq 0). \quad (8.3)$$

Consequently there is a probability measure $\mu_{d,t}$ supported on $[0, (1 + \sqrt{t})^{2d}]$ such that

$$H_{n,d}(t) = \int x^n d\mu_{d,t}(x). \quad (8.4)$$

Every leading Hankel determinant $\det(H_{i+j,d}(t))_{0 \leq i,j \leq r}$ is strictly positive. In particular, $H_{n,d}(t)^2 < H_{n-1,d}(t)H_{n+1,d}(t)$ for all $n \geq 1$.

Proof. Matrix multiplication expands (8.3) as a sum over the block-boundary paths of Section 6.1, with each vertical microstep of weight $\sqrt{t}$ and each horizontal one of weight $1+t$. For a returning path with j up and j down steps, the vertical weight is t^j , the same as before. This proves (8.3).

The Fourier multiplier of J is $1 + t + 2\sqrt{t}\cos\theta$, which lies between 0 and $(1 + \sqrt{t})^2$. Thus $0 \leq T_d \leq (1 + \sqrt{t})^{2d}I$. The spectral theorem applied to T_d and the unit vector e_0 gives (8.4) with the stated support and total mass.

The Hankel matrix is the Gram matrix of $e_0, T_d e_0, \dots, T_d^r e_0$. The vector $T_d^i e_0$ is supported on $\{0, \dots, di\}$ and has coefficient $t^{di/2} > 0$ at di . These vectors are therefore linearly independent by comparing their last nonzero coordinates. Their Gram determinant is strictly positive. In particular, the representing measure has infinite support. Applying strict Cauchy–Schwarz to $x^{(n-1)/2}$ and $x^{(n+1)/2}$ in this measure proves the strict log-convexity statement. $\square$

The shifted Hankel matrices are strictly positive definite as well. For a nonzero finitely supported v , Fourier transformation shows

$$\langle v, T_d v \rangle = \frac{1}{2\pi} \int_0^{2\pi} (1 + t + 2\sqrt{t} \cos \theta)^d |\widehat{v}(\theta)|^2 d\theta > 0.$$

The multiplier is positive except possibly at a single point when $t = 1$. Applying this to linear combinations of $T_d^i e_0$ proves the assertion. At $t = 0$, by contrast, $H_{n,d}(0) = 1$ and the Hankel matrices have rank one. No coefficientwise Hankel-positivity assertion in the indeterminate t is needed or claimed here.

9 Unequal block sizes

The factorization of Lemma 2.4 does not require every step to have the same block size. There is, however, an essential distinction between a *fixed order* of unequal sizes and a *sum over all orders*. This section states the positive result and the counterexample which shows that its qualification cannot be dropped. Throughout it, the budget equals the block size, so this generalizes the diagonal case $c = d$ only; the rectangular direction of Theorem 2.1 and the mixed-size direction here are orthogonal, and neither implies the other.

Let $\mathcal{D}$ be a finite set of positive integers. For a word $\kappa = (d_1, \dots, d_r)$ over $\mathcal{D}$, define $H_\kappa(t)$ by the prefix constraints at the partial sums $d_1 + \dots + d_i$, exactly as in (1.1); the empty word has polynomial 1. Introduce commuting variables x_d for $d \in \mathcal{D}$ and set

$$\mathcal{F}_\mathcal{D}(t; \mathbf{x}) = \sum_{\kappa \text{ a word over } \mathcal{D}} H_\kappa(t) \prod_{i=1}^{|\kappa|} x_{d_i}.$$

For a nonnegative multiplicity vector $\mathbf{m} = (m_d)_{d \in \mathcal{D}}$, put $|\mathbf{m}| = \sum_d m_d$ and $\|\mathbf{m}\| = \sum_d d m_d$.

Theorem 9.1 (Commutative mixed-size formula). *One has*

$$\boxed{\mathcal{F}_\mathcal{D}(t; \mathbf{x}) = \exp \left(\sum_{\mathbf{m} \neq 0} \frac{(|\mathbf{m}| - 1)!}{\prod_{d \in \mathcal{D}} m_d!} B_{\|\mathbf{m}\|}(t) \prod_{d \in \mathcal{D}} x_d^{m_d} \right)}. \quad (9.1)$$

For fixed multiplicities, the left side sums the polynomials of all distinct ordered words with those multiplicities.

Proof. Replace zS_d in Lemma 2.4 by $\mathcal{S}(u) = \sum_{d \in \mathcal{D}} x_d S_{d,d}(u, t)$; all formal operations are justified by positive total degree in the x_d . The same argument gives the exponential of $\left[\sum_{r \geq 1} \mathcal{S}^r / r \right]_{\geq 0}$. In a monomial with multiplicities $\mathbf{m}$, the multinomial coefficient is $|\mathbf{m}|! / \prod_d m_d!$, and by (3.2)

$$\prod_{d \in \mathcal{D}} S_{d,d}(u, t)^{m_d} = S_{\|\mathbf{m}\|, \|\mathbf{m}\|}(u, t).$$

Project to nonnegative powers, set $u = 1$, and apply Lemma 3.2 with $D = C = \|\mathbf{m}\|$. Dividing the multinomial coefficient by $|\mathbf{m}|$ gives (9.1). $\square$

It would be incorrect to infer that H_κ depends only on the multiset of its parts. Direct enumeration gives

$$\begin{aligned} H_{(1,1,2)}(t) &= 1 + 11t + 23t^2 + 11t^3 + t^4, \\ H_{(1,2,1)}(t) &= 1 + 11t + 22t^2 + 11t^3 + t^4, \\ H_{(2,1,1)}(t) &= 1 + 11t + 23t^2 + 11t^3 + t^4. \end{aligned} \quad (9.2)$$

The mixed formula gives their sum as the coefficient of $x_1^2 x_2$. This example also explains why the constant-block problem is special: a multiset of identical parts admits only one order. It is what makes the phrase “summed over orders” in Theorem 9.1 a genuine restriction rather than a formality.

Remark 9.2 (The reach of the scalar method). A fixed periodic pattern of unequal sizes may be grouped into a macrostep only if its internal block-boundary restrictions are retained; simply replacing the period by its total size erases inequalities and is generally invalid, as (9.2) shows. It does *not* follow that the method stops at constant block sizes. Theorem 9.1 shows that the same scalar factorization covers every finite set of block sizes, with the one qualification that its exponential computes the sum over all distinct orders with given multiplicities rather than any individual H_κ . What remains genuinely out of reach of the present argument is a closed formula for a single prescribed order of unequal sizes.

10 Singularity analysis and explicit leading asymptotics

Fix $d \geq 1$ and $t > 0$. Define

$$R(t) = (1 + \sqrt{t})^2, \quad \rho_d(t) = R(t)^{-d}, \quad K_d(t) = \prod_{r=0}^{d-1} g\left(\frac{\xi^r}{R(t)}, t\right). \quad (10.1)$$

The factors are obtained by continuation from their series at zero. In the factor with $r = 0$, the square root in (5.3) vanishes and the value is $R(t)/\sqrt{t}$. Conjugate factors pair, and $K_d(t)$ is a positive real number. It is also the finite boundary value $F_d(\rho_d(t), t)$, and admits a second, convergent expression

$$K_d(t) = \exp\left(\sum_{m \geq 1} \frac{B_{dm}(t)}{m R(t)^{dm}}\right). \quad (10.2)$$

Both are used below: the finite radical product of (10.1) is the fast evaluation, while the exponential sum (10.2) is what identifies the constant as the boundary value of F_d . Convergence of (10.2) follows from the coefficient asymptotics of the Narayana polynomial itself,

$$B_N(t) = \frac{1 + \sqrt{t}}{2\sqrt{\pi} t^{1/4}} R(t)^N N^{-1/2} (1 + O(N^{-1})), \quad (10.3)$$

which is the square-root transfer applied to (5.4) and makes the summands of (10.2) of order $m^{-3/2}$.

Theorem 10.1 (Uniform-chain asymptotics). *For fixed $d \geq 1$ and $t > 0$,*

$$H_{n,d}(t) = \frac{K_d(t)(1 + \sqrt{t})}{2\sqrt{\pi d} t^{1/4}} (1 + \sqrt{t})^{2dn} n^{-3/2} \left(1 + O(n^{-1})\right). \quad (10.4)$$

The radius of convergence of $F_d(\cdot, t)$ is $\rho_d(t)$, and $K_d(t) = F_d(\rho_d(t), t)$. In particular,

$$H_{n,d}(1) = \frac{K_d(1)}{\sqrt{\pi d}} 4^{dn} n^{-3/2} \left(1 + O(n^{-1})\right), \quad K_d(1) = \prod_{r=0}^{d-1} \left(\frac{2}{1 + \sqrt{1 - \xi^r}}\right)^2. \quad (10.5)$$

The square roots in the last formula are principal, with the $r = 0$ value zero. The error in (10.4) is locally uniform in a complex neighborhood of any fixed positive t , using the corresponding analytic branches; in particular it is locally uniform near $t = 1$.

Proof. The discriminant factors as

$$\Delta(w, t) = (1 - R(t)w)(1 - (1 - \sqrt{t})^2 w). \quad (10.6)$$

Its first positive root is $w_0 = 1/R(t)$; the other root is strictly farther from zero, or is absent when $t = 1$. There are no additional nonzero poles of (5.3): if its denominator vanished, multiplying it by the conjugate expression would give $4tw^2 = 0$, while at $w = 0$ the denominator is 2.

Write $w = w_0(1 - \varepsilon)$. Since

$$1 - \frac{(1 - \sqrt{t})^2}{(1 + \sqrt{t})^2} = \frac{4\sqrt{t}}{(1 + \sqrt{t})^2},$$

expanding the square root and denominator in (5.3) gives

$$g(w, t) = g(w_0, t) \left(1 - \frac{1 + \sqrt{t}}{t^{1/4}} \sqrt{\varepsilon} + O(\varepsilon) \right), \quad g(w_0, t) = \frac{R(t)}{\sqrt{t}}. \quad (10.7)$$

Choose the local branch $s = z^{1/d}$ with $s = w_0$ at $z = \rho_d(t)$. Only the factor $g(s, t)$ in (5.6) is singular there; all $g(\xi^r s, t)$ with $r \neq 0$ are analytic and nonzero. Moreover,

$$1 - s/w_0 = \frac{1}{d}(1 - z/\rho_d(t)) + O((1 - z/\rho_d(t))^2).$$

It follows that

$$F_d(z, t) = K_d(t) \left(1 - \frac{1 + \sqrt{t}}{t^{1/4}\sqrt{d}} \sqrt{1 - z/\rho_d(t)} + O(1 - z/\rho_d(t)) \right). \quad (10.8)$$

In fact the algebraic expression has a full convergent local Puiseux expansion, not just the displayed two terms. Its leading value is $K_d(t) > 0$, so the square-root term does not cancel, and it agrees with (10.2) by the convergent exponential formula.

To check dominance, a singularity of a factor at $\xi^r s = w_0$ always projects to the same point $z = w_0^d$. The second discriminant root projects outside this circle. Rotation invariance removes the apparent branching at $z = 0$. Thus there is a unique singularity on $|z| = \rho_d(t)$, with analytic continuation to a dented neighborhood beyond that circle. The same separation persists under sufficiently small complex perturbations of t .

Finally,

$$[z^n](1 - z/\rho)^{1/2} = -\frac{\rho^{-n}}{2\sqrt{\pi} n^{3/2}} (1 + O(n^{-1})).$$

The next nonanalytic Puiseux term has exponent $3/2$ and contributes $O(\rho^{-n} n^{-5/2})$; the locally analytic terms are handled by the continuation just established. This is the usual algebraic singularity transfer underlying directed-path asymptotics [4, 5]. Applying it to (10.8) proves (10.4), including the uniform form. Specialization $t = 1$ gives (10.5). $\square$

At $d = 1$, $K_1(1) = 4$, recovering the asymptotics of the shifted Catalan numbers. At $d = 2$,

$$K_2(1) = \frac{16}{(1 + \sqrt{2})^2} = 16(3 - 2\sqrt{2}) = 48 - 32\sqrt{2}, \quad H_{n,2}(1) \sim \frac{16(3 - 2\sqrt{2})}{\sqrt{2\pi}} 16^n n^{-3/2}. \quad (10.9)$$

d	$K_d(1)$	$K_d(1)/\sqrt{\pi d}$	growth factor 4^d
1	4.0000000000	2.2567583342	4
2	2.7451660041	1.0951627857	16
3	2.3004650571	0.7493420036	64
4	2.0653267193	0.5826179108	256
5	1.9170238358	0.4836905186	1024

Table 2: Decimal approximations to explicit algebraic boundary values and leading asymptotic constants. Numerical values are not interval-certified.

d	$K_d(1)$	ratio at $n = 100$	ratio at $n = 400$
1	4.0000000000	0.9743200738	0.9934736873
2	2.7451660041	0.9848655014	0.9961813732
3	2.3004650571	0.9886892314	0.9971536342
4	2.0653267193	0.9907300253	0.9976704816
8	1.6768915769	0.9941248072	0.9985270518

Table 3: Boundary constants and ratios of exact totals to the leading term in (10.5). Ratios approaching 1 illustrate, but do not prove, the asymptotic theorem.

The two displayed closed forms are the radical product and the value obtained from (10.2); they are the same number. The exponent $-3/2$ is characteristic of excursions; despite its exponential presentation, the generating function is algebraic rather than entire.

The asymptotic statement does not include $t = 0$. At that boundary $H_{n,d}(0) = 1$ and the square-root regime degenerates to a simple pole. Nor is any uniformity claimed when d grows with n , or when t tends to zero with n .

11 Support size: exact variance and a central limit theorem

Choose a lattice point uniformly from $Q_{n,d} \cap \mathbb{Z}^{dn}$ and let $X_{n,d}$ be its number of nonzero coordinates. Its probability generating function is $H_{n,d}(t)/H_{n,d}(1)$. Write $a_{n,d} = H_{n,d}(1)$.

Proposition 11.1 (Exact mean and variance). *For every $d \geq 1$ and $n \geq 0$,*

$$\mathbb{E}X_{n,d} = \frac{dn}{2}, \quad \boxed{\text{Var } X_{n,d} = \frac{d}{2a_{n,d}} \sum_{m=1}^n \binom{2dm-2}{dm-1} a_{n-m,d}}. \quad (11.1)$$

The empty sum gives variance zero when $n = 0$.

Proof. Palindromicity, that is $H_{n,d}(t) = t^{dn}H_{n,d}(t^{-1})$, makes the distribution symmetric about $dn/2$ and gives the exact mean. For the variance, introduce the centered series

$$G(z, s) = \sum_{n \geq 0} e^{-dns/2} H_{n,d}(e^s) z^n = F_d(ze^{-ds/2}, e^s).$$

By (1.5),

$$\log G(z, s) = \sum_{m \geq 1} \frac{z^m}{m} e^{-dms/2} B_{dm}(e^s).$$

Every centered polynomial on the right has first derivative zero at $s = 0$. For $N \geq 1$, Vandermonde's identity and its first two marked versions give

$$\sum_{j=0}^N \binom{N}{j}^2 (j - N/2)^2 = \binom{2N}{N} \frac{N^2}{4(2N-1)}. \quad (11.2)$$

For completeness, $B'_N(1) = N \binom{2N-1}{N-1}$ and $B''_N(1) = N(N-1) \binom{2N-2}{N-2}$; substituting these into $\sum j^2 \binom{N}{j}^2 - N \sum j \binom{N}{j}^2 + (N^2/4) B_N(1)$ proves (11.2), with $N = 1$ also immediate.

Since $\partial_s \log G(z, 0) = 0$, one has $\partial_s^2 G(z, 0) = G(z, 0) \partial_s^2 \log G(z, 0)$. When $N = dm$, the factor in (11.2) divided by m simplifies to

$$\frac{1}{m} \binom{2dm}{dm} \frac{(dm)^2}{4(2dm-1)} = \frac{d}{2} \binom{2dm-2}{dm-1}.$$

Taking the coefficient of z^n gives (11.1). □

Theorem 11.2 (Gaussian support fluctuations). *For fixed $d \geq 1$, as $n \rightarrow \infty$,*

$$\frac{X_{n,d} - dn/2}{\sqrt{dn/8}} \Rightarrow \mathcal{N}(0, 1), \quad \text{Var } X_{n,d} = \frac{dn}{8} + \kappa_d + O(n^{-1}), \quad (11.3)$$

where

$$\kappa_d = \frac{1}{16} + \frac{d}{8} - \frac{1}{8} \sum_{r=0}^{d-1} \sqrt{1 - \xi^r}. \quad (11.4)$$

The root sum is real; conjugate terms pair and the $r = 0$ term is zero. For example, $\kappa_1 = 3/16$ and $\kappa_2 = 5/16 - \sqrt{2}/8$.

Proof. Let $C_d(t)$ denote the leading prefactor in (10.4). Its locally uniform analytic form implies, for complex s near zero,

$$\log \mathbb{E} e^{sX_{n,d}} = n\lambda_d(s) + \log \frac{C_d(e^s)}{C_d(1)} + O(n^{-1}), \quad \lambda_d(s) = 2d \log \frac{1 + e^{s/2}}{2}. \quad (11.5)$$

Two independent justifications of this quasi-power step are available, and each suffices. The first is the locally uniform complex- t error statement of Theorem 10.1. The second tracks the distinguished branch point explicitly: choosing $\sqrt{t}$ analytic near $t = 1$, the branch point $1/R(t)$ varies analytically while the second root of (10.6) stays outside a larger disk, and the $d - 1$ nondistinguished factors of (5.6) remain analytic and nonzero on a common neighborhood of their evaluation points; hence the local Puiseux coefficients and the transfer estimates vary analytically and uniformly, and $C_d(t) = K_d(t)(1 + \sqrt{t})/(2\sqrt{\pi d} t^{1/4})$ is analytic and nonzero there. Either way the error and its derivatives are controlled on a smaller neighborhood by Cauchy's formula.

The derivatives of the rate at zero are $\lambda'_d(0) = d/2$ and $\lambda''_d(0) = d/8$. Substitution $s = iv/\sqrt{dn/8}$, followed by subtraction of the exact mean, makes (11.5) converge to $-v^2/2$; convergence of characteristic functions proves the central limit theorem. Twice differentiating gives the variance expansion, with $\kappa_d = (\log C_d(e^s))''|_{s=0}$.

It remains to evaluate this derivative. The non- K_d part of the prefactor is

$$\frac{1 + e^{s/2}}{2e^{s/4}\sqrt{\pi d}} = \frac{\cosh(s/4)}{\sqrt{\pi d}},$$

which contributes $1/16$. Put $v = \tanh(s/4)$ and $q = \xi^r$. At the critical argument $w = q/R(e^s)$, formula (5.3) becomes

$$g(w, e^s) = \frac{2}{1 - \frac{1}{2}(1 + v^2)q + \sqrt{1 - q}\sqrt{1 - v^2q}}. \quad (11.6)$$

At $v = 0$ the denominator is $(1 + \sqrt{1 - q})^2/2$. Its expansion in v^2 shows that the coefficient of v^2 in $\log g(w, e^s)$ is

$$\frac{q}{1 + \sqrt{1 - q}} = 1 - \sqrt{1 - q}.$$

Since $v^2 = s^2/16 + O(s^4)$, the second derivative is $(1 - \sqrt{1 - q})/8$. Summing over r and adding $1/16$ proves (11.4). $\square$

An equivalent positive, convergent series is

$$\kappa_d = \frac{1}{16} + \frac{d}{8} \sum_{m \geq 1} \frac{\binom{2dm}{dm}}{4^{dm}(2dm - 1)}. \quad (11.7)$$

To see this, expand $\sqrt{1 - z} = 1 - \sum_{k \geq 1} \binom{2k}{k} z^k / (4^k(2k - 1))$ and sum at the d th roots of unity. The series is absolutely convergent even at $z = 1$, so the filtering is justified. In particular, $\kappa_d > 1/16$.

For the double chain, the exact centered variance at $n = 400$ is approximately 0.13552348 after subtracting $dn/8 = 100$; the limiting correction is 0.13572330. The error is consistent with the proved $O(n^{-1})$ expansion. This numerical comparison is illustrative, whereas the constant is determined by the finite radical expression above.

This argument concerns the uniform distribution on *lattice points*, not uniform sampling of faces, of equivalence classes, or of support sets. Those would be different probability models.

12 Reproducibility, proof audit, and remaining questions

12.1 Independent checks actually performed

The package contains a single standard-library Python verifier which runs *two separate suites*, inherited from the two merged packages. They use different methods over disjoint parameter ranges, and neither range has been narrowed. All checks use arbitrary-precision integers or exact rational numbers, and the verifier raises explicit exceptions rather than relying on `assert`, so its checks remain active under `python -O`.

Suite 1 (rectangular). Three structurally different methods compute the same polynomials. The first performs dynamic programming on the original coordinates, merging states with equal current total and support size; it never uses the bridge formula. The second uses the decorated subset-walk macrosteps of (2.6), and is the only computational witness of the encoding of Lemma 2.2. The third uses the exponential-formula recurrence (4.3) with exact divisibility checks. All three agree for the 315 parameter triples

$$1 \leq d \leq 5, \quad 0 \leq c \leq 6, \quad 0 \leq n \leq 8.$$

The same run visits 34689 feasible arrays individually in 48 smaller cases, without merged states, and matches their support distributions. It verifies 225 instances of rectangular duality, 54 independent gamma path expansions, and 54 exact variance identities against (11.1). It also checks the quartic through z^{16} and matches the first four double-chain polynomials printed in the source.

Suite 2 (diagonal, deeper). The second suite computes $H_{n,d}$ from the block-weight slack dynamic program of (3.1), which uses the defining inequalities directly, and compares it with the recurrence over $1 \leq d \leq 8$, $0 \leq n \leq 12$: 104 complete polynomials, well beyond the range of Suite 1. It re-enumerates every individual coordinate vector in the 28 cases with $dn \leq 8$; verifies the endpoint-refined factorization (3.5) in 112 cases with $1 \leq d \leq 4$, $0 \leq n \leq 6$ and $t \in \{0, 1, 2, 7\}$ – the only computational witness of Corollary 3.1; reconstructs the gamma basis on all 104 rows and checks strict positivity; confirms Theorem 7.1 by finite differences in 40 cases ($1 \leq d \leq 8$, $1 \leq j \leq 5$), checking both the exact degree $2j$ and the leading coefficient (7.1); checks (7.6) and (7.7) on all 104 rows; re-checks the quartic through z^{16} coefficientwise in t ; and verifies the mixed-size identity (9.1) over all 121 ordered compositions with parts in $\{1, 2, 3\}$ and at most four blocks, against 35 multiplicity vectors, recording the counterexample (9.2).

The optional SymPy scripts derive the quartic by two different resultants, independently of the recurrence; the second reports that its difference from the printed quartic is the zero polynomial rather than a truncated series. The optional mpmath script computes exact integer totals, and exact rational means and variances, before any decimal conversion, then compares them with the asymptotic formulas at 80-digit working precision. It records observations up to $n = 400$ for $d \in \{1, 2, 3, 4, 5, 8\}$, the exact-to-asymptotic ratios of Table 3, the κ_d comparisons, and samples at the positive support weight $t = 4$. Those decimal calculations are not certified interval computations.

Finite checks corroborate the proofs; they do not certify untested parameters.

12.2 A short audit of the mathematical dependencies

Both proofs of the selected identity use only formal commutative series algebra together with an elementary count: the first uses the subset bijection and the binomial theorem, the second uses the slack process and stars and bars. Their only infinite objects are formal power series and Laurent series with exponents bounded above, handled coefficientwise. No appeal is made to convergence or numerical extrapolation. The source’s normalization is removed explicitly in (1.4). The main formal results are independent of both the analytic consequences and the computer calculations.

The algebraic formulas follow from a quadratic equation and a finite root-of-unity filter. The degree bound is an upper bound, proved twice, and the quartic has its formal branch specified by a nonzero derivative. The moment representation uses the spectral theorem for a bounded positive operator. The asymptotic and limit-law sections additionally use analytic continuation, square-root singularity transfer [5], and the continuity theorem for characteristic functions; their hypotheses are checked in place. These analytic conclusions are separate from, and are not needed to prove, the original conjectural identity.

Particular pitfalls which are avoided are: imposing bounds inside a block; identifying a preorder-equivalence block with an antichain; confusing support h with Ehrhart h^* ; counting strict rather than weak excursions; using an additive instead of multiplicative root-of-unity filter; replacing the compressed power PJ^dP by the power of a compression; substituting $u = 1$ into the slack symbol before projecting; and assuming that a cyclic orbit has exactly one nonnegative rotation.

12.3 Reproduction commands

From the package directory, the complete exact check is

```
python code/verify.py
```

It needs only Python 3.10 or later and no third-party package. By default it writes to `results/`; `-output DIRECTORY` redirects it. All generated coefficient arrays use increasing powers. The optional symbolic and numerical checks are

```
python code/derive_quartic.py
python code/symbolic_certificate.py
python code/asymptotics.py
```

The first two require SymPy and the third mpmath; `asymptotics.py` additionally accepts `-maximum N` (default 400). Versions used in this run are recorded in `results/environment.json`. The article can be rebuilt with

```
latexmk -pdf -interaction=nonstopmode article.tex
```

or with the two `pdflatex` passes performed by `build.sh` and by the optional `Makefile`. The original supplied manifest is preserved in `context/manifest.tex`; the two merged packages shipped byte-identical copies of it, and only one is kept. The proposed additional entry in `manifest-entry.tex` describes this package without modifying that source file.

12.4 What would be a genuinely different next problem?

The all-block-size identity selected here no longer has an unresolved case within either proof, and Theorem 9.1 extends the method to arbitrary finite sets of block sizes once one sums over orders. What remains open is of a different kind. A closed formula for a single prescribed order of unequal sizes is not delivered by the present scalar factorization, and (9.2) shows that no formula depending only on the multiset can exist. Real-rootedness of the support polynomials, the minimal algebraic degree for a given d (only the upper bound 2^d is proved here), gamma positivity for arbitrary preorders, and the weighted q -reciprocity conjecture of the same source all require additional arguments. Finally, a direct bijection explaining the permutation average (4.5), with all periodicity factors treated explicitly, would be a useful complement to the formal proofs.

A A compact proof certificate

The logical core of the second proof can be recorded in six equations, so that the main result can be audited without reading any consequence section. All projections act in $\mathbb{Q}[t]((u^{-1}))[[z]]$:

$$\begin{aligned} S_{d,c}(u, t) &= u^c \left(\frac{u+t-1}{u-1} \right)^d, & M &= 1 + z [S_{d,c}M]_{\geq 0}, \\ L &= -\log(1 - zS_{d,c}) = L_+ + L_-, & M_* &= \exp(L_+), \\ (1 - zS_{d,c})M_* &= \exp(-L_-), & [\exp(-L_-)]_{\geq 0} &= 1. \end{aligned}$$

The recurrence uniquely gives $M = M_*$. Since

$$\left[S_{d,c}^m \right]_{\geq 0} \Big|_{u=1} = \sum_{a_1 + \dots + a_{dm} \leq cm} t^{\#\{i: a_i > 0\}} = \sum_{r \geq 0} \binom{dm}{r} \binom{cm}{r} t^r,$$

specializing $u = 1$ proves (2.2), and $c = d$ gives the conjectured formula. The only infinite series in u have exponents bounded above, and the specialization is made only after projection, coefficient by coefficient in z .

B Exact data and source provenance

File	Contents
<code>results/verification.json</code>	Exact test ranges, counts of cases, and pass status for both verifier suites.
<code>results/exact_polynomials.json</code>	All 315 rectangular support polynomials in increasing coefficient order; the only data with $c \neq d$.
<code>results/coefficients.json</code>	Every diagonal polynomial for $1 \leq d \leq 8$, $0 \leq n \leq 12$, with its gamma coefficients; the only data reaching $d = 8$ and $n = 12$.
<code>results/gamma_polynomials.json</code>	Gamma coefficients obtained by independent uncolored microstep dynamic programming.
<code>results/small_values.csv</code>	Uniform-chain totals, support coefficients, and exact rational variances.
<code>results/lattice_point_totals.csv</code>	Totals at $t = 1$ over the deeper diagonal range.
<code>results/quartic_certificate.txt</code>	The first symbolic resultant, the printed quartic, and the branch check.
<code>results/symbolic_certificate.json</code>	The second elimination certificate, with a zero-polynomial residual.
<code>results/asymptotics.csv</code>	Exact-count versus asymptotic ratios and variance-correction comparisons.
<code>results/asymptotic_constants.csv</code> , <code>results/asymptotic_checks.csv</code> <code>results/constants.json</code>	Boundary constants and exact-to-leading-term ratios at $n = 100$ and $n = 400$, including $d = 8$. Decimal values of $K_d(1)$, leading constants, and κ_d for $1 \leq d \leq 5$; the κ_d column occurs nowhere else.
<code>results/pdf_validation.json</code>	Machine-readable render audit of the compiled article.
<code>results/environment.json</code>	Exact Python, SymPy, mpmath, and pdfTeX versions.
<code>sources.md</code> , <code>STATUS.md</code>	Source/version audit and explicit claim boundaries.

The target was checked against the text of Section 9.5, printed page 24 of the arXiv v1 PDF, and the corresponding HTML formula. The mathematical formula was read in structured HTML and parsed PDF text; repeated attempts to obtain a screenshot of that page failed with retrieval and cache errors, and no unavailable screenshot is treated as evidence. The source paper is not included in this archive; retrieval details are recorded in `sources.md`. The supplied manifest is included unaltered, with its hash recorded in `sources.md`.

References

- [1] C. A. Athanasiadis and F. Chapoton, *Polytopes and posets associated to preorders*, arXiv:2605.26916v1 (2026), especially Section 9.5, p. 24. [arXiv record](#); [HTML text](#). Accessed 20 September 2026.

- [2] C. A. Athanasiadis, *Lattice point enumeration of polytopes associated to integer compositions*, arXiv:2510.23903v1 (2025), Equation (1), Theorem 1.2 and Section 2. [arXiv record](#). Accessed 20 September 2026.
- [3] F. Spitzer, A combinatorial lemma and its application to probability theory, *Transactions of the American Mathematical Society* **82** (1956), 323–339. [doi:10.1090/S0002-9947-1956-0079851-X](#).
- [4] C. Banderier and P. Flajolet, Basic analytic combinatorics of directed lattice paths, *Theoretical Computer Science* **281** (2002), 37–80. [doi:10.1016/S0304-3975\(02\)00007-5](#).
- [5] P. Flajolet and A. Odlyzko, Singularity analysis of generating functions, *SIAM Journal on Discrete Mathematics* **3** (1990), no. 2, 216–240. [doi:10.1137/0403019](#).
- [6] Z. Dai, Q. Hou, Z. Liu, W. Thawinrak, and H. Wang, *Counting Lattice Points in Minkowski Sums of Cross Polytopes*, arXiv:2608.16037v2 (2026). [HTML text](#). Consulted as adjacent preorder-support and duality literature; its introduction connects support duality to Conjecture 5.4 of [1].
- [7] X. Yan, *Variations of colored multiset Eulerian polynomials and applications*, arXiv:2608.15682v2 (2026). [HTML text](#). Consulted as adjacent composition-polytope literature.
- [8] V. Reshetnikov, *manifest.tex*, user-supplied L^AT_EX research catalogue of seventy-one packages, received 20 September 2026. The catalogue records claimed results, not independent proof verification. A copy of the exact supplied file is included in `context/manifest.tex`.